

FIRE FIGHTING RESCUERS EYE BOT

Abstract—This research paper explores the emerging field of autonomous rescue bots and their potential impact on firefighting operations. With the aim of enhancing rescue missions and minimizing risks for both firefighters and victims, autonomous rescue bots offer a promising solution in navigating hazardous environments and providing vital assistance. The paper begins by examining the capabilities of these bots, including mobility, sensory perception, communication systems, and decision-making algorithms. It then evaluates their effectiveness in real-world scenarios, comparing their performance to human-led operations. The challenges and limitations in implementing rescue bots, such as technical, regulatory, and ethical issues, are identified and discussed. Furthermore, the implications of adopting autonomous rescue bots in terms of firefighter safety, resource allocation, and operational efficiency are explored. The paper concludes by discussing potential future advancements, such as the integration of artificial intelligence and machine learning, to further enhance the capabilities of these bots. Overall, this research aims to contribute to safer and more effective firefighting practices by shedding light on the potential of autonomous rescue bots.

Index Terms— fire fighting robot, rescue robot, obstacle avoiding robot, ESP32-CAM project

I. INTRODUCTION

In the face of escalating challenges posed by modern urban fires, firefighting agencies are constantly seeking innovative solutions to improve rescue operations and mitigate the risks faced by both firefighters and victims. The emergence of autonomous robotic systems has sparked considerable interest as a potential game-changer in firefighting scenarios. Among these systems, autonomous rescue bots stand out as promising tools that could revolutionize the way rescue missions are executed. Traditionally, firefighting operations have heavily relied on human intervention, exposing firefighters to hazardous environments and life-threatening situations. The deployment of autonomous rescue bots aims to address this issue by providing an additional layer of protection, reducing human exposure to danger, and enhancing overall mission effectiveness. These bots are specifically designed to navigate through complex and hazardous environments, locate and rescue victims, and assist firefighters in critical tasks.

- To lessen the life risks of human firefighter.
- To detect fire and extinguish the fire.
- To detect person and to do a live video stream of the incident happened place.

II. RELATED WORKS

Our work has maintain an integrity among more than 3 projects. In this projects the developers has done wonderful work but consisting only one specified feature. Our team has brought their works together and made those specified projects a multi-functional featured robot. The works and papers that helped us making this variation. Ninad Mehandele [1] in their project they just detect objects and human using OpenCV and ESP 32 cam module. Gomathy et al. proposed a system [2] that can detect different colour using ESP 32 cam

and OpenCV, but we detect fire using the colour detection concept as fire has three basic colour (red, orange and yellow). Patil et al. [3] created a home security system using ESP 32 cam. In their project when the LDR sensor module was chosen as the data input for the activation ESP 32 cam, then the ESP 32 cam will take a photo and upload it to Google Drive. Vairavan et al. [4] as well as we have worked for obstacle avoiding car, but our car has fire detection feature added to that. Rai et al. [5] has done work for Bluetooth Controlled Car, we also have done this work with speaking enable mode. Shah et al. [6] project has worked for fire detection and extinguishing, but at the same time our project work for obstacle avoiding also.

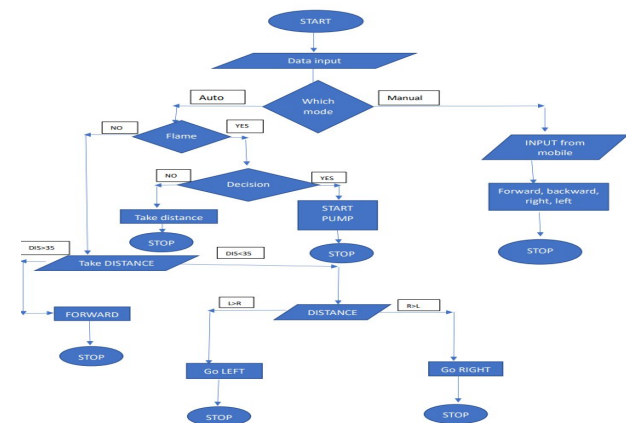


Fig. 1. Working flow chart.

III. METHODOLOGY

The main objective of our project is to design and build a robot that can perform two different tasks: controlling the fire and finding the injured person in a disaster scenario. The robot has two modes of operation that correspond to these tasks.

A. Fire Fighting Mode

This mode is activated when the robot detects a fire in its vicinity. The robot uses an ultrasonic sensor to measure the distance to the nearest obstacle and avoid colliding with it. The robot also uses an IR flame sensor to determine the direction and intensity of the fire. The robot navigates towards the area where the distance to the obstacle is the maximum, and stops when it reaches the fire zone. Then, it turns the water pump on but the source to the water body is connected to different place to make a safety use of our sensor used module. Then, that source completely put out the fire. The robot also has an ESP 32 cam module that captures video data and sends it to a device, where a machine learning algorithm processes the data and detects the fire regions in the images.

B. Monitoring Mode

This mode is activated when the robot needs to find and help the injured people in the disaster area. We use a Bluetooth module (HC 06) to communicate with the robot and control its movement through our input. The robot also uses the ESP 32 cam module to take pictures of its surroundings and compare them with a database of human faces. If the robot finds a match, it means that it has found a person. The robot then takes a photo of the person and uploads it to Google Drive, where we can access it and provide assistance. If the robot does not find a match, it means that there is no person in that area. The robot then continues to move forward and search for other people.

C. Circuit Diagram and Components

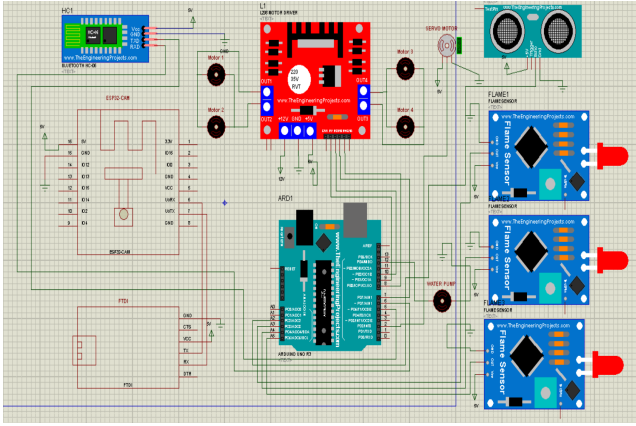


Fig. 2. Circuit connection of the robot.

- **ARDUINO UNO:** Arduino Uno is a microcontroller board based on the ATmega328P. It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a 16 MHz ceramic resonator, a USB connection, a power jack, an ICSP header and a reset button.
- **ULTRASONIC SENSOR:** Ultrasonic sensors are electronic devices that calculate the target's distance by emission of ultrasonic sound waves and convert those waves into electrical signals. There are mainly two essential elements: the transmitter and receiver. Using piezoelectric crystals, the transmitter generates sound, and it travels to the target and returns to the receiver. The calculation is given as $D = 1/2 T \times C$, where T corresponds to time measured in seconds and C corresponds to sound speed = 343 m/s.
- **L298N MOTOR DRIVER:** A motor driver acts as an interface between the motor and the microcontroller because the microcontroller and motor work on different voltage and current ranges. Motor driver ICs are commonly connected to the motor controller through an H-bridge circuit.
- **FLAME SENSOR:** The flame sensor module is used in flame alarms and detects flame wavelengths within the range of 760 nm–1100 nm from the light source. The sensor can be damaged by high temperature and should therefore be placed at a suitable distance from the flame. The paper reports flame detection from a 100 cm distance and a detection angle of 60°.
- **ESP32-CAM:** The ESP32-based camera module developed by AI-Thinker uses a 32-bit CPU with combined Wi-Fi and Bluetooth/BLE. It has 520 KB SRAM with external 4M PSRAM. Its GPIO pins support UART, SPI, I2C, PWM, ADC and DAC. The module combines with the OV2640 camera, with a reported maximum camera resolution up to 1600 × 1200. It supports an SD card up to 4 GB for captured images and provides a compact, cost-effective camera and wireless platform.
- **SERVO MOTOR:** The SG90 Micro Servo Motor operates on 4.8–6 VDC (5 V typical) and can rotate 180 degrees. The paper reports approximately 10 mA at idle and 100–250 mA while moving. The servo is connected to a PWM-enabled pin.
- **BLUETOOTH MODULE:** The paper describes the HM-06/HC-06 Bluetooth module as a short-range wireless communication module using Bluetooth 2.0. It is described as a slave device and as using frequency-hopping spread spectrum (FHSS). The reported operating frequency range is 2.402–2.480 GHz.

IV. RESULT

Mode	Attempts	Worked	Accuracy (%)
Obstacle Avoiding	50	47	93
Firefighting Mode	50	45	90
Human Detection Mode	25	23	92

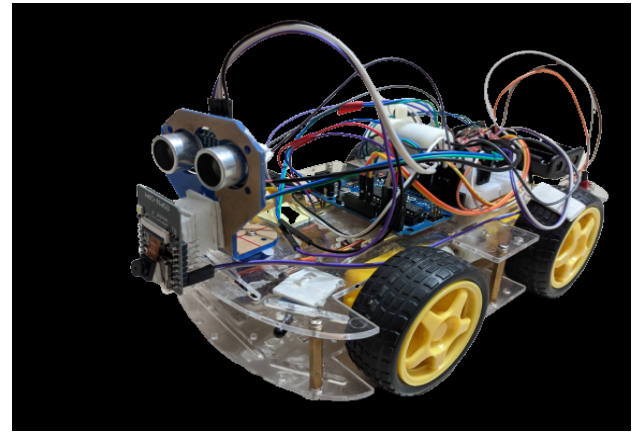


Fig. 3. Picture of the robot.

V. FUTURE WORK

A. Emergency Response in Buildings

During a fire incident in a building, the robot can navigate through smoke-filled environments, assess the situation, and relay critical information to firefighters. It can detect fire sources, locate trapped individuals, and provide real-time updates on the structural integrity of the building.

B. Industrial Fires and Hazardous Environments

In industrial settings where fires or hazardous material incidents occur, the robot can be deployed to assess the situation before human responders enter. It can gather data on the type and severity of the incident, identify potential risks, and provide essential information to guide firefighting and containment efforts.

C. Search and Rescue Missions

The robot's capabilities make it valuable for search and rescue missions, such as in natural disasters or urban emergencies. It can navigate through rubble or inaccessible areas, detect signs of life using its sensors and image processing capabilities, and relay accurate victim locations to rescue teams.

D. Wildfire Monitoring and Suppression

Deploying the robot in wildfire-prone regions can aid in early detection and monitoring of fires. It can autonomously patrol designated areas, detect fire outbreaks, and alert authorities. Additionally, equipped with fire suppression mechanisms, it can assist in extinguishing small fires before they escalate.

E. Hazardous Material Detection and Handling

The robot's sensors, including the flame sensor and gas sensors, enable it to detect and identify hazardous materials or leaks. It can be utilized to inspect storage facilities, chemical plants, or nuclear sites, helping to identify potential risks and assisting in containment and mitigation efforts.

F. Post-disaster Assessment

Following a disaster, such as an earthquake or flood, the robot can be deployed to assess the structural integrity of buildings or infrastructure. It can navigate through unstable environments, capture visual data, and transmit it to authorities to aid in damage assessment and post-disaster recovery planning.

G. Remote Surveillance and Monitoring

The robot can serve as a remote surveillance device in various settings, such as monitoring high-security areas, public

events, or remote locations. It can provide real-time video feeds and sensor data to enhance situational awareness and support security personnel in making informed decisions. It is important to adapt and customize the robot's functionality based on specific application requirements. By integrating advanced technologies and constantly improving its capabilities, this robot can play a vital role in ensuring the safety of both responders and the public in various emergency situations.

VI. DISCUSSION

A. Comparison

[1], [2], [3] has individually working sense of object detection and human detection, colour detection concept of fire's 3 colours, image uploading to Google Drive but in our project we combine all these 3 projects and make 1 project. [4] has only Obstacle avoiding mode but our has many features like fighting the fire while detecting fire. [5] has only autonomous fire fighting mode but our robot has both manual and autonomous.

VII. CONCLUSION

In conclusion, the development of this advanced robot for fire-fighting and rescue missions represents a powerful blend of technology and compassion. Its capabilities to detect fires, navigate hazardous environments, locate victims, and communicate crucial information have the potential to save lives and minimize risks for responders. By harnessing the power of automation and artificial intelligence, this robot serves as a powerful tool in our ongoing efforts to protect and rescue those in need. It stands as a testament to our commitment to leveraging technology for the greater good, reminding us that even in the face of adversity, innovation and empathy can work hand in hand to create a safer and more compassionate world.

REFERENCES

- [1] Mehendale, N. (2022). Object Detection using ESP 32 CAM. Available at SSRN.
- [2] Gomathy, D. C., CHAITANYA, M. C., ANVITHA, M. B. J. S., JAHNAVI, M. C. (2021). Color Detection Using Pandas and Opencv. International Journal of Scientific Research in Engineering and Management (IJSREM), Volume 5.
- [3] Mah, Y. W., Mokhtar, M. H. (2023). IoT Application for the Development of Home Security Systems. Progress in Engineering Application and Technology, 4(1), 459–467.
- [4] Vairavan, R., Kumar, S. A., Ashiff, L. S., Jose, C. G. (2018). Obstacle avoidance robotic vehicle using ultrasonic sensor, Arduino controller. International Research Journal of Engineering and Technology (IRJET), 5(02).
- [5] Rai, N., Rasaily, D., Wangchuk, T. R., Gurung, M., Khawas, R. K. (2016). Bluetooth remote controlled car using Arduino. Int. J. Eng. Trends Technol, 33, 381–384.
- [6] Shah, S. S., Shah, V. K., Mamtara, P., Hapani, M. (2013). Fire fighting robot. Int. J. Emerg. Trends Technol. Comp. Appl, 2(4), 232–234.
- [7] Amit, Y., Felzenszwalb, P., Girshick, R. (2020). Object detection. Computer Vision: A Reference Guide, 1–9.
- [8] Amano, H. (2002, August). Present status and problems of fire fighting robots. In Proceedings of the 41st SICE Annual Conference. SICE 2002. (Vol. 2, pp. 880–885). IEEE.
- [9] Chaudhry, A., Batra, M., Gupta, P., Lamba, S., Gupta, S. (2019, October). Arduino Based Voice Controlled Robot. In 2019 International Conference on Computing, Communication, and Intelligent Systems (ICCCIS) (pp. 415–417). IEEE.
- [10] Baras, N., Nantzios, G., Ziouzos, D., Dasygenis, M. (2019, May). Autonomous obstacle avoidance vehicle using lidar and an embedded

system. In 2019 8th International Conference on Modern Circuits and Systems Technologies (MOCAST) (pp. 1–4). IEEE.

[11] Chen, T. H., Wu, P. H., Chiou, Y. C. (2004, October). An early fire-detection method based on image processing. In 2004 International Conference on Image Processing, 2004. ICIP'04. (Vol. 3, pp. 1707–1710). IEEE.